Lab 8

Practice 1

```
student@Echo2:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nol
ogin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/
nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nolog
in
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologi
n
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aper
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```
student@Echo2:~$ tac /etc/passwd
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexist
ent:/usr/sbin/nologin
student:x:1000:1000:KevinG,,,:/home/student:/bin/bash
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/us
r/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
saned:x:107:111::/var/lib/saned:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-r
emote-desktop:/usr/sbin/nologin
geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/s
bin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexisten
t:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:
/bin/false
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologi
n
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nolog
in
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/
nologin
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nol
```


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```

student@Echo2: ~

2: student@Echo2: ~

```
root:x:0:0:root:/root:/bin/bash
student@Echo2:~$ cat -nE /etc/passwd
     1  root:x:0:0:root:/root:/bin/bash$
     2  daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin$
     3  bin:x:2:2:bin:/bin:/usr/sbin/nologin$
     4  sys:x:3:3:sys:/dev:/usr/sbin/nologin$
     5  sync:x:4:65534:sync:/bin:/bin/sync$
     6  games:x:5:60:games:/usr/games:/usr/sbin/nologin$
     7  man:x:6:12:man:/var/cache/man:/usr/sbin/nologin$
     8  lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin$
     9  mail:x:8:8:mail:/var/mail:/usr/sbin/nologin$
    10  news:x:9:9:news:/var/spool/news:/usr/sbin/nologin$
    11  uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin$
    12  proxy:x:13:13:proxy:/bin:/usr/sbin/nologin$
    13  www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin$
    14  backup:x:34:34:backup:/var/backups:/usr/sbin/nologin$
    15  list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologi
n$
    16  irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin$
    17  _apt:x:42:65534::/nonexistent:/usr/sbin/nologin$
    18  nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin$
    19  systemd-network:x:998:998:systemd Network Management:/:/usr/
sbin/nologin$
    20  dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/f
alse$
    21  tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false$
    22  systemd-timesync:x:991:991:systemd Time Synchronization:/:/u
sr/sbin/nologin$
    23  messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sb
in/nologin$
    24  dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin$
```

1: student@Echo2: ~

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```

student@Echo2: ~

2: student@Echo2: ~

```
    32  saned:x:107:111::/var/lib/saned:/usr/sbin/nologin$
    33  polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin$
    34  rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin$
    35  colord:x:109:113:colord colour management daemon:/var/lib/co
lord:/usr/sbin/nologin$
    36  Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bi
n/false$
    37  student:x:1000:1000:KevinG,,,:/home/student:/bin/bash$
    38  _flatpak:x:111:115:Flatpak system-wide installation helper:/
nonexistent:/usr/sbin/nologin$
student@Echo2:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
student@Echo2:~$ tail -n 5 /etc/passwd
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:109:113:colord colour management daemon:/var/lib/colord:/us
r/sbin/nologin
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
student:x:1000:1000:KevinG,,,:/home/student:/bin/bash
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexist
ent:/usr/sbin/nologin
student@Echo2:~$ tail -1 /etc/passwd
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexist
ent:/usr/sbin/nologin
student@Echo2:~$ head -1 /etc/passwd
root:x:0:0:root:/root:/bin/bash
student@Echo2:~$
```

1: student@Echo2: ~

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aper
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```

Practice 2

1/1 student@Echo2: ~

```
student@Echo2:~$ cut -d ":" -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
```

1/1 student@Echo2: ~

```
student@Echo2:~$ cut -d ":" -f1,6 /etc/passwd
root:/root
daemon:/usr/sbin
bin:/bin
sys:/dev
sync:/bin
games:/usr/games
man:/var/cache/man
lp:/var/spool/lpd
mail:/var/mail
news:/var/spool/news
uucp:/var/spool/uucp
proxy:/bin
www-data:/var/www
backup:/var/backups
list:/var/list
irc:/run/ircd
_apt:/nonexistent
nobody:/nonexistent
systemd-network:/
dhcpcd:/usr/lib/dhcpcd
tss:/var/lib/tpm
systemd-timesync:/
messagebus:/nonexistent
dnsmasq:/var/lib/misc
avahi:/run/avahi-daemon
speech-dispatcher:/run/speech-dispatcher
usbmux:/var/lib/usbmux
cups-pk-helper:/nonexistent
fwupd-refresh:/var/lib/fwupd
geoclue:/var/lib/geoclue
gnome-remote-desktop:/var/lib/gnome-remote-desktop
```

1/1 student@Echo2: ~

```
student@Echo2:~$ cut -d ":" -f1,6,7 --output-delimiter=" > " /etc/passwd
root > /root > /bin/bash
daemon > /usr/sbin > /usr/sbin/nologin
bin > /bin > /usr/sbin/nologin
sys > /dev > /usr/sbin/nologin
sync > /bin > /bin/sync
games > /usr/games > /usr/sbin/nologin
man > /var/cache/man > /usr/sbin/nologin
lp > /var/spool/lpd > /usr/sbin/nologin
mail > /var/mail > /usr/sbin/nologin
news > /var/spool/news > /usr/sbin/nologin
uucp > /var/spool/uucp > /usr/sbin/nologin
proxy > /bin > /usr/sbin/nologin
www-data > /var/www > /usr/sbin/nologin
backup > /var/backups > /usr/sbin/nologin
list > /var/list > /usr/sbin/nologin
irc > /run/ircd > /usr/sbin/nologin
_apt > /nonexistent > /usr/sbin/nologin
nobody > /nonexistent > /usr/sbin/nologin
systemd-network > / > /usr/sbin/nologin
dhcpcd > /usr/lib/dhcpcd > /bin/false
tss > /var/lib/tpm > /bin/false
systemd-timesync > / > /usr/sbin/nologin
messagebus > /nonexistent > /usr/sbin/nologin
dnsmasq > /var/lib/misc > /usr/sbin/nologin
avahi > /run/avahi-daemon > /usr/sbin/nologin
speech-dispatcher > /run/speech-dispatcher > /bin/false
usbmux > /var/lib/usbmux > /usr/sbin/nologin
cups-pk-helper > /nonexistent > /usr/sbin/nologin
fwupd-refresh > /var/lib/fwupd > /usr/sbin/nologin
geoclue > /var/lib/geoclue > /usr/sbin/nologin
gnome-remote-desktop > /var/lib/gnome-remote-desktop > /usr/sbin/nologin
```

1/1 student@Echo2: ~

```
student@Echo2:~$ cut -d ";" -f1 ~/Documents/sample_files/Csv/cereal.csv
name
String
100% Bran
100% Natural Bran
All-Bran
All-Bran with Extra Fiber
Almond Delight
Apple Cinnamon Cheerios
Apple Jacks
Basic 4
Bran Chex
Bran Flakes
Cap'n'Crunch
Cheerios
Cinnamon Toast Crunch
Clusters
Cocoa Puffs
Corn Chex
Corn Flakes
Corn Pops
Count Chocula
Cracklin' Oat Bran
Cream of Wheat (Quick)
Crispix
Crispy Wheat & Raisins
Double Chex
Froot Loops
Frosted Flakes
Frosted Mini-Wheats
Fruit & Fibre Dates, Walnuts, and Oats
Fruitful Bran
```

```
1/1                                student@Echo2: ~
Triples
Trix
Wheat Chex
Wheaties
student@Echo2:~$ cut -d ";" -f1,4 --output-delimiter=" calories = "~/Docucuments/sample_files/Csv/cereal.csv


^Z
[1]+  Stopped                 cut -d ";" -f1,4 --output-delimiter=" calories = "~/Docucuments/sample_files/Csv/cereal.csv
student@Echo2:~$ cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv

^Z
[2]+  Stopped                 cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv
student@Echo2:~$ cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv
^Z
[3]+  Stopped                 cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv
student@Echo2:~$ cut -d";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv

^Z
student@Echo2:~$ cut -d";" -f1,4 --output-delimiter=" calories = "~Documents/sample_files/Csv/cereal.csv
^Z
[5]+  Stopped                 cut -d";" -f1,4 --output-delimiter=" calories = "~Documents/sample_files/Csv/cereal.csv
student@Echo2:~$ cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv
^Z
[6]+  Stopped                 cut -d ";" -f1,4 --output-delimiter=" calories = "~/Documents/sample_files/Csv/cereal.csv
student@Echo2:~$ cut -d";" -f1,4 --output-delimiter=" calories = " ~/Documents/sample_files/Csv/cereal.csv
name calories = calories
String calories = Int
100% Bran calories = 70
100% Natural Bran calories = 120
All-Bran calories = 70
All-Bran with Extra Fiber calories = 50
Almond Delight calories = 110
```

Practice 3

```
1/1                                student@Echo2: ~/Documents/sample_files/Csv
student@Echo2:~/Documents/sample_files/Csv$ sort cars.csv
AMC Ambassador Brougham;13.0;8;360.0;175.0;3821.;11.0;73;US
AMC Ambassador DPL;15.0;8;390.0;190.0;3850.;8.5;70;US
AMC Ambassador SST;17.0;8;304.0;150.0;3672.;11.5;72;US
AMC Concord;19.4;6;232.0;90.00;3210.;17.2;78;US
AMC Concord;24.3;4;151.0;90.00;3003.;20.1;80;US
AMC Concord d/l;18.1;6;258.0;120.0;3410.;15.1;78;US
AMC Concord DL;23.0;4;151.0;0;3035.;20.5;82;US
AMC Concord DL 6;20.2;6;232.0;90.00;3265.;18.2;79;US
AMC Gremlin;18.0;6;232.0;100.0;2789.;15.0;73;US
AMC Gremlin;19.0;6;232.0;100.0;2634.;13.0;71;US
AMC Gremlin;20.0;6;232.0;100.0;2914.;16.0;75;US
AMC Gremlin;21.0;6;199.0;90.00;2648.;15.0;70;US
AMC Hornet;18.0;6;199.0;97.00;2774.;15.5;70;US
AMC Hornet;18.0;6;232.0;100.0;2945.;16.0;73;US
AMC Hornet;19.0;6;232.0;100.0;2901.;16.0;74;US
AMC Hornet;22.5;6;232.0;90.00;3085.;17.6;76;US
AMC Hornet Sportabout (sw);18.0;6;258.0;110.0;2962.;13.5;71;US
AMC Matador;14.0;8;304.0;150.0;3672.;11.5;73;US
AMC Matador;15.0;6;258.0;110.0;3730.;19.0;75;US
AMC Matador;15.5;8;304.0;120.0;3962.;13.9;76;US
AMC Matador;16.0;6;258.0;110.0;3632.;18.0;74;US
AMC Matador;18.0;6;232.0;100.0;3288.;15.5;71;US
AMC Matador (sw);14.0;8;304.0;150.0;4257.;15.5;74;US
AMC Matador (sw);15.0;8;304.0;150.0;3892.;12.5;72;US
AMC Pacer;19.0;6;232.0;90.00;3211.;17.0;75;US
AMC Pacer d/l;17.5;6;258.0;95.00;3193.;17.8;76;US
AMC Rebel SST;16.0;8;304.0;150.0;3433.;12.0;70;US
AMC Rebel SST (sw);0;8;360.0;175.0;3850.;11.0;70;US
AMC Spirit DL;27.4;4;121.0;80.00;2670.;15.0;79;US
Audi 100LS;20.0;4;114.0;91.00;2582.;14.0;73;Europe
Audi 100LS;23.0;4;115.0;95.00;2694.;15.0;75;Europe
```

```
1/1                                student@Echo2: ~/Documents/sample_files/Csv
Volkswagen Dasher;30.5;4;97.00;78.00;2190.;14.1;77;Europe
Volkswagen Dasher (diesel);43.4;4;90.00;48.00;2335.;23.7;80;Europe
Volkswagen Jetta;33.0;4;105.0;74.00;2190.;14.2;81;Europe
Volkswagen Model 111;27.0;4;97.00;60.00;1834.;19.0;71;Europe
Volkswagen Pickup;44.0;4;97.00;52.00;2130.;24.6;82;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.0;75;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.2;76;Europe
Volkswagen Rabbit;29.5;4;97.00;71.00;1825.;12.2;76;Europe
Volkswagen Rabbit;29.8;4;89.00;62.00;1845.;15.3;80;Europe
Volkswagen Rabbit;41.5;4;98.00;76.00;2144.;14.7;80;Europe
Volkswagen Rabbit C (Diesel);44.3;4;90.00;48.00;2085.;21.7;80;Europe
Volkswagen Rabbit Custom;29.0;4;97.00;78.00;1940.;14.5;77;Europe
Volkswagen Rabbit Custom;31.9;4;89.00;71.00;1925.;14.0;79;Europe
Volkswagen Rabbit Custom Diesel;43.1;4;90.00;48.00;1985.;21.5;78;Europe
Volkswagen Rabbit l;36.0;4;105.0;74.00;1980.;15.3;82;Europe
Volkswagen Scirocco;31.5;4;89.00;71.00;1990.;14.9;78;Europe
Volkswagen Super Beetle 117;0;4;97.00;48.00;1978.;20.0;71;Europe
Volkswagen Super Beetle;26.0;4;97.00;46.00;1950.;21.0;73;Europe
Volkswagen Type 3;23.0;4;97.00;54.00;2254.;23.5;72;Europe
Volvo 144ea;19.0;4;121.0;112.0;2868.;15.5;73;Europe
Volvo 145e (sw);18.0;4;121.0;112.0;2933.;14.5;72;Europe
Volvo 244DL;22.0;4;121.0;98.00;2945.;14.5;75;Europe
Volvo 245;20.0;4;130.0;102.0;3150.;15.7;76;Europe
Volvo 264gl;17.0;6;163.0;125.0;3140.;13.6;78;Europe
Volvo Diesel;30.7;6;145.0;76.00;3160.;19.6;81;Europe
student@Echo2:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
student@Echo2:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
408 sorted_cars.csv
student@Echo2:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv
22663 sorted_cars.csv
student@Echo2:~/Documents/sample_files/Csv$
```

Question 2

```
1/1                                student@Echo2: ~
student@Echo2:~$ pwd
/home/student
student@Echo2:~$ wc -l access.log
100 access.log
student@Echo2:~$ head -n 1 access.log
2025-05-16 07:28:11 Walter /logout SUCCESS
student@Echo2:~$ tail -n 1 access.log
2025-12-21 07:59:16 Julia /upload SUCCESS
student@Echo2:~$ man head
student@Echo2:~$ head -v -n 15 access.log
==> access.log <==
2025-05-16 07:28:11 Walter /logout SUCCESS
2025-08-11 16:31:30 Honami /logout DENIED
2025-09-27 00:04:47 Ken /upload DENIED
2025-10-15 12:24:38 Honami /login SUCCESS
2025-06-03 15:46:56 Asimov /404 SUCCESS
2025-11-03 00:09:48 Shinji /admin DENIED
2025-01-04 21:27:55 Rize /404 FAIL
2025-09-25 12:51:21 Honami /settings SUCCESS
2025-08-13 10:56:49 Jet /dashboard FAIL
2025-12-19 06:38:40 Edward /logout SUCCESS
2025-08-17 05:07:16 Shinji /dashboard DENIED
2025-10-02 22:51:56 Honami /upload DENIED
2025-08-05 01:13:11 Asimov /404 FAIL
2025-01-03 11:40:57 Kei /login FAIL
2025-11-12 02:12:43 Rize /admin FAIL
student@Echo2:~$ tails -v -n 15 access.log
bash: tails: command not found
student@Echo2:~$ tail -v -n 15 access.log
==> access.log <==
2025-08-26 02:35:31 Jet /settings SUCCESS
2025-11-08 22:34:21 Shinji /admin SUCCESS
```

```
student@Echo2:~$ cut -d ' ' -f3 access.log
Walter
Honami
Ken
Honami
Asimov
Shinji
Rize
Honami
Jet
Edward
Shinji
Honami
Asimov
Kei
Rize
Shinji
Touka
Kei
Annie
Ken
Ein
Rize
Eren
Edward
Rize
Edward
Arisu
Rize
Shinji
Spike
Honami
```

```
Julia
student@Echo2:~$ man sort
student@Echo2:~$ sort -k3, 3 access.log -o sorted_access_log_by_uname.log
sort: invalid number after ',': invalid count at start of ''
student@Echo2:~$ sort -k3,3 access.log -o sorted_access_log_by_uname.log
student@Echo2:~$ sort -k4,4 access.log -o sorted_access_log_by_endpoint.log
student@Echo2:~$ cut -d' ' -f3,4 --output-delimiter=' accessed: ' access.log
Walter accessed: /logout
Honami accessed: /logout
Ken accessed: /upload
Honami accessed: /login
Asimov accessed: /404
Shinji accessed: /admin
Rize accessed: /404
Honami accessed: /settings
Jet accessed: /dashboard
Edward accessed: /logout
Shinji accessed: /dashboard
Honami accessed: /upload
Asimov accessed: /404
Kei accessed: /login
Rize accessed: /admin
Shinji accessed: /dashboard
Touka accessed: /404
Kei accessed: /admin
Annie accessed: /logout
Ken accessed: /logout
Ein accessed: /login
Rize accessed: /settings
Eren accessed: /404
Edward accessed: /reset
Rize accessed: /admin
```